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METHOD FOR FORMING A HOLDER SURFACE, AND GRINDING APPARATUS

Abstract

A method for forming a holder surface of a chuck table by rotating a spindle, to which an annular grinding stone is attached, and grinding an upper surface of the chuck table with the grinding stone rotated along with the spindle. The chuck table includes a porous member and a frame having a dense body in an annular shape. The method includes a first grinding process including tilting an axis of the chuck table with respect to an axis of the spindle relatively at a predetermined first angle and grinding an entire upper surface of the frame with the grinding stone, and a second grinding process including tilting the axis of the chuck table with respect to the axis of the spindle relatively at a second angle, which is smaller than the first angle, and grinding an entire upper surface of the porous member with the grinding stone.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-025224 filed on Jun. 22, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a method for forming a holder surface to hold a wafer and to a grinding apparatus.

BACKGROUND

[0003] Japanese Patent Laid-Open Publication No. **2018-114573** discloses a method for grinding a wafer held on a conical chuck table with a lower surface of an annular grinding stone on radial range basis. Moreover, Japanese Patent Laid-Open Publication No. 2014-237210 discloses a so-called self-grinding method for grinding an upper surface of a replacing new chuck table with a replacing new grinding wheel to form a holder surface of the chuck table and a grinding surface of the grinding wheel to be parallel to each other. The chuck table may be composed of a porous member that forms the holder surface and a dense frame that surrounds the porous member annularly.

SUMMARY

[0004] The porous member and the frame composing the chuck table may have different degree of hardness; therefore, as the surface of the chuck table is processed through the self-grinding operation, grinding of the porous member may progress ahead compared to grinding of the frame. As a result of this self-grinding operation, a ring-shaped concavity may be formed on an outer periphery of the porous member at a boundary between the porous member and the frame. Accordingly, a problem may occur such that a wafer held on this chuck table and ground with this grinding wheel may have a slight convexity in a ring shape on an outer periphery corresponding to the position of the ring-shaped concavity of the porous member.

[0005] In view of such a problem, one of the purposes of the present disclosure is to provide a method for forming a holder surface and a grinding apparatus that may prevent a porous member from a ring-shaped concavity to be formed on an outer periphery thereof during self-grinding of a chuck table.

[0006] According to an aspect of the present disclosure, a method for forming a holder surface of a chuck table to suction and hold a wafer thereon by rotating a spindle, to which an annular grinding stone is attached, and grinding an upper surface of the chuck table with the grinding stone rotated along with the spindle, is provided. The chuck table is composed of a porous member and a frame having a dense body in an annular shape surrounding an outer circumference of the porous member. The method includes a first grinding process including tilting an axis of the chuck table with respect to an axis of the spindle relatively at a predetermined first angle and grinding an entire upper surface of the frame with the grinding stone, and a second grinding process including tilting the axis of the chuck table with respect to the axis of the spindle relatively at a second angle, which is smaller than the first angle, and grinding an entire upper surface of the porous member with the grinding stone.

[0007] A grinding apparatus configured to form the holder surface of the chuck table in the method is provided. The grinding apparatus includes the chuck table configured to hold the wafer, a table spindle configured to rotate the chuck table, a grinding unit configured to grind the wafer held on the chuck table with the grinding stone attached to the spindle, a tilt-adjusting assembly configured

to adjust relative inclination between the table spindle and the spindle to one of the first angle or the second angle, and a controller configured to control the tilt-adjusting assembly to form the first angle for grinding the frame with the grinding stone and to form the second angle for grinding the porous member with the grinding stone.

[0008] According to the present disclosure, in the first grinding process and the second grinding process, in which the grain diameters of the grinding stones are different, the relative angle between the axis of the spindle and the axis of the chuck table is changed, and areas in the chuck table to be ground are changed between the first grinding stone and the second grinding stone as set forth above. Accordingly, the porous member may prevent a ring-shaped concavity, which may otherwise be formed in self-grinding due to progress of grinding of the porous member ahead compared to the frame, from being formed on a circumferential portion thereof. As a result, after self-grinding, when the wafer is held and ground on the chuck table, the wafer may prevent convexity corresponding to the form of ring-shaped concavity from being formed thereon, and the wafer may be ground to an even thickness.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a schematic perspective view of the grinding apparatus according to an embodiment of the present disclosure.

[0010] FIG. 2 is a side view of a chuck table, a table-rotating assembly, and a tilt-adjusting assembly.

[0011] FIG. 3 is a schematic perspective view of the chuck table, the table-rotating assembly, and the tilt-adjusting assembly.

[0012] FIG. 4 is a cross-sectional partial view of the chuck table and a table spindle.

[0013] FIG. 5 is a cross-sectional plan view of an exemplary arrangement of the tilt-adjusting assembly.

[0014] FIG. 6 illustrates a preparatory grinding process.

[0015] FIG. 7 illustrates a first grinding process.

[0016] FIG. 8 illustrates a second grinding process.

DESCRIPTION OF EMBODIMENTS

[0017] A grinding apparatus according to an embodiment of the present disclosure will be described below with reference to the accompanying drawings. FIG. 1 is a schematic perspective view of the grinding apparatus according to the embodiment of the present disclosure.

[0018] As shown in FIG. 1, a grinding apparatus 1 may grind and process a wafer W, which is a workpiece to be processed, formed in a substantially circular shape of a disk. The grinding apparatus 1 is configured to perform a series of operations including a loading process, a grinding process, a cleaning process, and an unloading process with the wafer W fully automatically. To load the grinding apparatus 1 with the wafer W, a cassette 13 containing the wafer W is set in the grinding apparatus 1.

[0019] In the grinding apparatus 1, an X-axis direction, a Y-axis direction, and a Z-axis direction are orthogonal to one another. The X-axis direction and the Y-axis direction are substantially horizontal directions, and the Z-axis direction is an up-down direction (vertical direction). A +X side and a -X side, which are one and the opposite sides as pointed by a bidirectional arrow along the X-axis direction, are a frontward side and a rearward side, respectively. A +Y side and a -Y side, which are one and the opposite sides as pointed by a bidirectional arrow along the Y-axis direction, are a leftward side and a rightward side, respectively. A +Z side and a -Z side, which are one and the opposite sides as pointed by a bidirectional arrow along the Z-axis direction, is an upper side and a lower side, respectively.

[0020] On the frontward side to a base **10** of the grinding apparatus **1**, two cassettes **13** that may each store a plurality of wafers **W** are placed. One of the cassettes **13** may store the wafers **W** before grinding, and the other of the cassettes **13** may store the wafers **W** after grinding. At a rearward position with respect to the cassettes **13**, a robot hand **11** that may pick up the wafer **W** from one of the cassettes **13** or may place the wafer **W** back in the other of the cassettes **13** is provided.

[0021] At a rightward-rear position and a leftward-rear position with respect to the robot hand **11**, respectively, provided are a positioning assembly **21**, on which the wafer **W** before grinding may be placed, and a spinner-cleaning device **12**, which may clean the wafer **W** after grinding. The robot hand **11** may convey the wafer **W** before grinding from the cassette **13** to the positioning assembly **21** and convey the wafer **W** after grinding from the spinner-cleaning device **12** to the cassette **13**.

[0022] The positioning assembly **21** includes a plurality of positioning pins **23**, which are arranged around a temporary placement table **22** and are movable inward toward or outward from a center of the temporary placement table **22**. With the wafer **W** placed on the temporary placement table **22** in the positioning assembly **21**, by moving the plurality of positioning pins **23** to respectively contact an outer circumference of the wafer **W**, a center of the wafer **W** is located at the center of the temporary placement table **22**.

[0023] The spinner-cleaning device **12** may clean the wafer **W** having been ground by supplying a cleaning solution from a cleaning nozzle **121** to the wafer **W** and rotating a spinner table **122** holding the wafer **W** at a high speed by a motor, which is not shown.

[0024] Between the positioning assembly **21** and the spinner-cleaning device **12** in the Y-axis direction, provided are a first conveyer assembly **15**, which may convey the wafer **W** before grinding from the positioning assembly **21** to a chuck table **14**, and a second conveyer assembly **16**, which may convey the wafer **W** after grinding from the chuck table **14** to the spinner-cleaning device **12**. The first conveyer assembly **15** and the second conveyer assembly **16** each have a conveyer pad, which suctions and holds the wafer **W** from above, and a supporting arm supporting the conveyer pad. The first conveyer assembly **15** and the second conveyer assembly **16** may convey the wafer **W** by operating the supporting arm supporting the conveyer pad to rotate on an axis extending in the Z-axis direction and move up or down in the Z-axis direction.

[0025] On an upper surface of the base **10** on the rearward side in the grinding apparatus **1**, an opening in a rectangular form elongated in the X-axis direction is formed. The opening is covered by a movable plate **17**, which is movable in the X-axis direction along with the chuck table **14**, and a waterproof cover **18** having a form of bellows.

[0026] Below the waterproof cover **18**, provided is a table-movable assembly **20** that may move the chuck table **14** in the X-axis direction. The table-movable assembly **20** includes a pair of guide rails **25** extending in the X-axis direction and a ball screw **26**, and a movable table **27** is supported on the guide rails **25** movably along the guide rails **25**. The ball screw **26** is screwed to a screw thread (not shown) in the movable table **27**, and when the ball screw **26** is rotated by a motor **24**, the movable table **27** moves in the X-axis direction.

[0027] Below the waterproof cover **18**, provided are a table-rotating assembly **30**, which may rotate and drive the chuck table **14**, and a tilt-adjusting assembly **36**, which may adjust inclination of the chuck table **14**. The table-rotating assembly **30** and the tilt-adjusting assembly **36** will be described further below.

[0028] The chuck table **14** is composed of a disk-shaped porous member **42** and a frame **43** including an annular portion that surrounds an outer periphery of the porous member **42**.

[0029] The porous member **42** is made of a porous material and has fine pores formed throughout. An upper surface of the porous member **42** forms a holder surface **421**. The holder surface **421** suctions and thereby holds the wafer **W** by a suction source, which is not shown. The frame **43** is formed of ceramics and is composed of a dense body, of which hardness is higher than the porous member **42** made of the porous material. An upper surface of the chuck table **14** is formed as a

conical surface, of which apex is located on an axis C1 being a rotation axis of the chuck table 14, and which descends gradually toward an outer periphery of the chuck table 14 (see FIG. 6). Note that, in FIG. 6, inclination of the holder surface 421 with respect to the horizontal direction and inclination of the axis C1 with respect to the vertical direction are depicted in an exaggerated manner; however, the inclinations in reality are slight enough not to be visually recognizable. [0030] Next, a horizontal movable assembly 50, lift/lower assemblies 60, and a grinding assembly 70 will be described. The horizontal movable assembly 50 is located on a frontward side or a column 19, which stands at a rearward position on the base 10. On the horizontal movable assembly 50, the grinding assembly 70 is supported via two lift/lower assemblies 60.

[0031] The horizontal movable assembly 50 includes a pair of guide rails 51 arranged on the frontward side of the column 19 and extending in the Y-axis direction, movable tables 52 arranged on the pair of guide rails 51 movably in the Y-axis direction, and a ball screw 53 extending in the Y-axis direction and screwed to a screw thread (not shown) formed in the movable tables 52. The two movable tables 52 are arranged side by side along the Y-axis direction. The movable tables 52 are movable in the Y-axis direction by rotation of the ball screw 53, which is rotatable by a driving force of a motor 54 coupled to an end of the ball screw 53.

[0032] The two lift/lower assemblies 60 are similarly configured and are each arranged on a frontward side of a respective one of the movable tables 52. Each lift/lower assembly 60 has a pair of guide rails 61 arranged on the frontward side of the movable table 52 and extending in the Z-axis direction. Furthermore, the lift/lower assembly 60 has a lift/lower table 62, which is arranged on the pair of guide rails 61 movably in the Z-axis direction, and a ball screw 63 extending in the Z-axis direction and screwed to a screw thread (not shown) formed in the lift/lower table 62. The lift/lower table 62 are movable in the Z-axis direction by rotation of the ball screw 63, which is rotatable by a driving force of a motor 64 coupled to one end of the ball screw 63.

[0033] The grinding assembly 70 includes a first grinding unit (grinding unit) 71 supported by one of the lift/lower assemblies 60 on the right (-Y side) and a second grinding unit (grinding unit) 72 supported by the other of the lift/lower assemblies 60 on the left (+Y side). The grinding assembly 70 may grind an upper surface of the wafer W suctioned and held on the holder surface 421 of the chuck table 14 with a first grinding stone (grinding stone) 80 and a second grinding stone (grinding stone) 90, which will be described further below.

[0034] The first grinding unit 71 is mounted on a frontward side of the lift/lower table 62 via a first holder 74 and supports a first spindle (spindle) 76 rotatably with respect to a first spindle housing 75 supported by the first holder 74. The first spindle 76 may rotate on an axis C21, which is parallel to the Z-axis direction (see FIGS. 6 and 7) by a driving force of a first spindle motor 77.

[0035] The first spindle 76 is coupled with a first mount 78 at a lower end thereof, and to the first mount 78, a first grinding wheel 79 is attached. On a lower surface of the first grinding wheel 79, the first grinding stone 80, including a plurality of first grinding stones 80 arranged annularly, is provided. As such, the plurality of first grinding stones 80 are attached to the first spindle 76 via the first mount 78 and the first grinding wheel 79. The first grinding stones 80 may be, for example, grinding stones that are usable in rough grinding, and a size of abrasive grains contained therein is relatively large. In other words, the first grinding unit 71 that may grind the workpiece with the first grinding stones 80 may function as a rough-grinding assembly that may grind the wafer W roughly.

[0036] The second grinding unit 72 is mounted on a frontward side of the lift/lower table 62 via a second holder 84 and supports a second spindle (spindle) 86 rotatably with respect to a second spindle housing 85 supported by the second holder 84. The second spindle 86 may rotate on an axis C22 (see FIG. 8), which is parallel to the Z-axis direction by a driving force of a second spindle motor 87.

[0037] The second spindle 86 is coupled with a second mount 88 at a lower end thereof, and to the second mount 88, a second grinding wheel 89 is attached. On a lower surface of the second

grinding wheel **89**, the second grinding stone **90**, including a plurality of second grinding stones **90** arranged annularly, is provided. As such, the plurality of second grinding stones **90** are attached to the second spindle **86** via the second mount **88** and the second grinding wheel **89**. The second grinding stones **90** may be, for example, grinding stones that are usable in finish grinding, and a size of abrasive grains contained therein is relatively small, and a grain diameter of the second grinding stones is smaller than a grain diameter of the first grinding stones **80**. In other words, the second grinding unit **72** that may grind the workpiece with the second grinding stones **90** may function as a finish-grinding assembly that may finely grind to finish the wafer W.

[0038] On the base **10**, a thickness-measuring device **45** is provided. The thickness-measuring device **45** includes a first height gauge for measuring a height position of the upper surface of the wafer W held on the holder surface **421** of the chuck table **14** and a second height gauge for measuring a height position of the upper surface of the chuck table **14**. Based on a difference between a value measured with the first height gauge and a value measured with the second height gauge, a thickness of the wafer W is measured.

[0039] Operations of the components in the grinding apparatus **1** are controlled by a controller **95**. The controller **95** includes a processor that may execute various processes and a memory (memory) storing various parameters and programs. The memory of the controller **95** may store, as parts of the control programs, for example, programs for controlling the operations of the table-movable assembly **20**, the table-rotating assembly **30**, the tilt-adjusting assembly **36**, the horizontal movable assembly **50**, the lift/lower assemblies **60**, and the grinding assembly **70**. For the operation of each component in the grinding apparatus **1** described below, when no explicit entity is specified as a subject to control the operation, it is assumed that the operation is controlled by signals output from the controller **95**.

[0040] FIG. **2** is a side view of the chuck table **14**, the table-rotating assembly **30**, and the tilt-adjusting assembly **36**. As shown in FIG. **2**, the table-rotating assembly **30** that may drive the chuck table **14** to rotate includes a cylindrical table spindle **31**, of which axis C**1** is common with the chuck table **14**, located below the frame **43**.

[0041] FIG. **3** is a schematic perspective view of the chuck table **14**, the table-rotating assembly **30**, and the tilt-adjusting assembly **36**. As shown in FIG. **3**, the table-rotating assembly **30** further includes a motor **32**, a belt pulley **33** provided to an output shaft of the motor **32**, and a transmission belt **34** wound around the belt pulley **33** and the table spindle **31**. When the motor **32** drives the belt pulley **33** to rotate, the rotational force is transmitted to the table spindle **31** via the transmission belt **34**, and the table spindle **31** having the same axis C**1** as the chuck table **14** rotates.

[0042] The tilt-adjusting assembly **36** is provided to adjust inclination of the table spindle **31** and chuck table **14**. The tilt-adjusting assembly **36** includes a support stand **37**, position-adjusting units **38**, and a fixed support **39** connected to the support stand **37**.

[0043] The support stand **37** includes a cylindrically-shaped support cylinder **371** and a disc-shaped flange **372**, which is a lower part of the support cylinder **371** expanded radially. The tilt-adjusting assembly **36** adjusts the inclination of the table spindle **31** by operating the position-adjusting units **38** to tilt the flange **372** using the fixed support **39** as a fulcrum.

[0044] FIG. **4** is a cross-sectional partial view of the chuck table **14** and the table spindle **31**. As shown in FIG. **4**, the table spindle **31** is inserted in the support cylinder **371** of the support stand **37**. A bearing **373** arranged inside the support cylinder **371** is in contact with an outer circumferential surface of the table spindle **31**, and the table spindle **31** is rotatably supported through the bearing **373**.

[0045] Referring back to FIG. **3**, at different positions in a circumferential direction of the support stand **37**, at least two position-adjusting units **38** are provided and are connected to the flange **372**. Each position-adjusting unit **38** includes a cylinder **381** fixed to the movable table **27**, a movable shaft **382** penetrating the cylinder **381**, a motor **383** connected to a lower end of the movable shaft **382**, a male screw (not shown) formed at a tip end of the movable shaft **382**, a female screw (not

shown) formed on the flange **372** to be screwed with the male screw, and a pressurizer **384** located on the flange **372** and having a spring that provides backlash between the male screw and the female screw.

[0046] The cylinder **381** is fitted in a hole formed through the movable table **27** in the Z-axis direction. In the position-adjusting unit **38**, as the motor **383** rotates and drives the movable shaft **382**, a height position of the flange **372**, in which the female screw is formed, may change in the Z-axis direction.

[0047] FIG. **5** is a cross-sectional plan view of an exemplary arrangement of the position-adjusting units **38** and the fixed support **39**. According to the configuration shown in FIG. **5**, two position-adjusting units **38** and one fixing support **39** are arranged at intervals by **120** degrees (equally spaced) in the circumferential direction. The fixed support **39** supports the flange **372** at a constant height position. The two position-adjusting units **38** may operate individually to change the height position of the flange **372**.

[0048] Next, an overall operation of the grinding apparatus **1** will be explained briefly. As shown in FIG. **1**, in the grinding apparatus **1**, an unground wafer **W** is picked up from the cassette **13** and conveyed to the positioning assembly **21** by the robot hand **11**. Next, the upper surface of the wafer **W** is suctioned and held by the conveyer pad of the first conveyer assembly **15**, and the wafer **W** is conveyed from the positioning assembly **21** to the chuck table **14** by the first conveyer assembly **15**. For passing the wafer **W** from the conveyer pad of the first conveyer assembly **15** to the chuck table **14**, the chuck table **14** is driven by the table-movable assembly **20** and located at a load/unload position in proximity to (frontward in the X-axis direction) the first conveyer assembly **15**.

[0049] When the wafer **W** is passed to the chuck table **14** by the first conveyer assembly **15**, the wafer **W** is suctioned and held on the holder surface **421** of the chuck table **14**. Thereafter, the chuck table **14** is moved rearward by the table-movable assembly **20**, and the wafer **W** is located below the grinding assembly **70**.

[0050] In the grinding assembly **70**, the horizontal movable assembly **50** locates the first grinding unit **71** above the chuck table **14**, and the table-rotating assembly **30** rotates the chuck table **14** and the wafer **W** being held and suctioned thereon. The lift/lower assembly **60** lowers the first grinding unit **71** and rotates the first grinding stones **80** through the first spindle **76**. As such, the first grinding stones **80** and the wafer **W** on the chuck table **14** are moved to rotate respectively and contact each other, thereby grinding the upper surface of the wafer **W** roughly.

[0051] After grinding the wafer **W** roughly, the horizontal movable assembly **50** locates the second grinding unit **72** above the chuck table **14**, and the table-rotating assembly **30** continuously rotates the chuck table **14** and the wafer **W** held and suctioned thereon. Moreover, the lift/lower assembly **60** lowers the second grinding unit **72** and rotates the second grinding stones **90** through the second spindle **86**. As such, the second grinding stones **90** and the wafer **W** on the chuck table **14** are moved to rotate respectively and contact each other, thereby grinding to finish the upper surface of the wafer **W**, and grinding of the wafer **W** is completed.

[0052] After grinding is completed, the table-movable assembly **20** is operated to move the chuck table **14** to return to the load/unload position. The wafer **W** having been ground is conveyed by the second conveyer assembly **16** to the spinner-cleaning device **12** and cleaned thereby. After cleaning, the wafer **W** having been ground is stored in the cassette **13** by the robot hand **11**.

[0053] According to the grinding apparatus **1** described herein, a process called self-grinding to preparatorily grind the upper surface of the chuck table **14** is performed. Self-grinding may be performed, for example, after replacing the first and/or second grinding wheels **79**, **89** or after grinding the wafers **W** a predetermined number of times so that the holder surface **421** of the chuck table **14** is formed to be parallel to the lower surfaces (grinding surfaces) of the first and second grinding stones **80**, **90**. The following describes a method for forming the holder surface **421** of the chuck table **14** by self-grinding according to the present embodiment.

[0054] The method for forming the holder surface **421** according to the present embodiment is implemented by performing steps including a preparatory grinding process, a first grinding process, and a second grinding process in this given order. FIG. **6** illustrates the preparatory grinding process, FIG. **7** illustrates the first grinding process, and FIG. **8** illustrates the second grinding process. According to the present embodiment, in each of the processes, the axis **C21** of the first spindle **76** and the axis **C22** of the second spindle **86** are maintained parallel to the vertical direction (Z-axis direction), but the axis **C1** of the chuck table **14** (table spindle **31**) is reoriented.

Preparatory Grinding Process

[0055] As shown in FIG. **6**, in the preparatory grinding process, the controller **95** (see FIG. **1**) controls and drives the tilt-adjusting assembly **36** (see FIG. **2**) so that the axis **C1** of the chuck table **14** is tilted relatively to the axis **C21** of the first spindle **76** at a preparation angle $\theta 0$. Under this condition, the table-movable assembly **20** and the horizontal movable assembly **50** are driven to locate the chuck table **14** below the first grinding unit **71**. At this position, outer circumferential edges of the annularly arranged first grinding stones **80** are maintained to pass directly above the axis **C1** of the chuck table **14**. The preparation angle $\theta 0$ is a relative inclination between the table spindle **31** and the first spindle **76**.

[0056] Thereafter, the table-rotating assembly **30** is operated to rotate the chuck table **14** on the axis **C1**, and the first spindle **76** is driven to rotate the first grinding stones **80** on the axis **C21**.

Moreover, the lift/lower assembly **60** is driven to lower the first grinding unit **71** to press the rotating first grinding stones **80** against the upper surface of the rotating chuck table **14**. Thereby, the holder surface **421** being the upper surface of the porous member **42** in the chuck table **14** and the upper surface of the frame **43** are mutually ground and form the identical conical surfaces.

First Grinding Process

[0057] After completion of the preparatory grinding process, the first grinding process is performed. As shown in FIG. **7**, in the first grinding process, the controller **95** controls and drives the tilt-adjusting assembly **36** so that the axis **C1** of the chuck table **14** is tilted relatively to the axis **C21** of the first spindle **76** at a first angle $\theta 1$. The first angle $\theta 1$ is a relative inclination between the table spindle **31** and the first spindle **76**. The first angle $\theta 1$ is set to be greater than the preparation angle $\theta 0$ by a predetermined angle. Under this condition, similarly to the preparatory grinding process, the chuck table **14** is located below the first grinding unit **71**, and the outer circumferential edges of the annularly arranged first grinding stones **80** are maintained to pass directly above the axis **C1** of the chuck table **14**.

[0058] Thereafter, the table-rotating assembly **30** is operated to rotate the chuck table **14** on the axis **C1**, and the first spindle **76** is driven to rotate the first grinding stones **80** on the axis **C21**.

Moreover, the lift/lower assembly **60** is driven to lower the first grinding unit **71** to press the rotating first grinding stones **80** against the upper surface of the rotating chuck table **14** to grind the upper surface of the chuck table **14**.

[0059] In this grinding process, the first angle $\theta 1$ is set to be greater than the preparation angle $\theta 0$; therefore, a contact area between the first grinding stones **80** and the upper surface of the chuck table **14** is changed compared to the contact area in the preparation grinding process. In particular, a part of the first grinding stones **80** closer to the axis **C21** is caused to contact the entire upper surface of the frame **43** of the chuck table **14**. In other words, the first angle $\theta 1$ is set to an angle that enables the first grinding stones **80** to contact, within the chuck table **14**, solely the upper surface of the frame **43** but not to contact the upper surface (holder surface **421**) of the porous member **42**.

[0060] By tilting the chuck table **14** as described above, the entire upper surface of the frame **43** is ground, but the porous member **42** is not ground; therefore, the upper surface of the frame **43** and the upper surface of the porous member **42** are formed to have respective conical surfaces, of which taper angles are different. More specifically, the upper surface of the frame **43** is formed to have a taper angle smaller than a taper angle of the upper surface of the porous member **42**.

Second Grinding Process

[0061] After completion of the first grinding process, the second grinding process is performed. As shown in FIG. 8, in the second grinding process, the controller 95 controls and drives the tilt-adjusting assembly 36 so that the axis C1 of the chuck table 14 is tilted relatively to the axis C22 of the second spindle 86 at a second angle $\theta 2$. The second angle $\theta 2$ is a relative inclination between the table spindle 31 and the second spindle 86. The second angle $\theta 2$ is set to be smaller than the first angle $\theta 1$ by a predetermined angle and to be equal to or smaller than the preparation angle $\theta 0$ by a predetermined angle. Under this condition, the chuck table 14 is located below the second grinding unit 72, and outer circumferential edges of the annularly arranged second grinding stones 90 are maintained to pass directly above the axis C1 of the chuck table 14.

[0062] Thereafter, the table-rotating assembly 30 is operated to rotate the chuck table 14 on the axis C1, and the second spindle 86 is driven to rotate the second grinding stones 90 on the axis C22. Moreover, the lift/lower assembly 60 is driven to lower the second grinding unit 72 to press the rotating second grinding stones 90 against the upper surface of the rotating chuck table 14 to grind the upper surface of the chuck table 14.

[0063] In this grinding process, the second angle $\theta 2$ is set to be smaller than the first angle $\theta 1$; therefore, a contact area between the second grinding stones 90 and the upper surface of the chuck table 14 is changed compared to the contact area between the first grinding stones 80 and the upper surface of the chuck table 14 in the first grinding process. In particular, a part of the second grinding stones 90, excluding a part closer to the axis C22, is caused to contact the entire upper surface of the porous member 42 in the chuck table 14. In other words, the second angle $\theta 2$ is set to an angle that enables the second grinding stones 90 to contact, within the chuck table 14, merely the upper surface of the porous member 42 (holder surface 421) but not to contact the upper surface of the frame 43.

[0064] In the second grinding process, which is after forming the upper surface of the frame 43 and the upper surface of the porous member 42 to have the respective conical surfaces at the different taper angles in the first grinding process, the chuck table 14 is ground in the orientation tilted as described above. As such, while the upper surface of the frame 43 is prevented from being ground, the holder surface 421 forming the entire upper surface of the porous member 42 may be ground to be finished with the second grinding stones 90.

[0065] According to the above embodiment, the angle of the axis C1 of the chuck table 14 is changed between the first grinding process and the second grinding process in the self-grinding operation. With the changed angles, the process to grind the frame 43 and the porous member 42 is separated into the first grinding process, in which the upper surface of the frame 43 is ground with the first grinding stones 80, and the second grinding process, in which the upper surface of the porous member 42 is ground with the second grinding stones 90. Note that, although the above embodiment includes the preparatory grinding process, the preparatory grinding process is not necessarily required.

[0066] Conventionally, the frame and the porous member are ground simultaneously in self-grinding; therefore, grinding of the porous member may progress ahead compared to grinding of the frame, resulting to form a ring-shaped concavity on the circumferential portion of the porous member.

[0067] In contrast, by separating the processes of grinding the frame 43 and the porous member 42 and performing in sequence, as those in the above embodiment, progression of grinding of the porous member 42 ahead compared to the frame 43 may be prevented, and the porous member 42 may prevent the ring-shaped concavity from being formed on the circumferential portion thereof. Accordingly, after self-grinding, when the wafer W is ground, the wafer W may prevent the convexity corresponding to the form of ring-shaped concavity from being formed thereon, and the wafer W may be ground to an even thickness.

[0068] Moreover, as the porous member 42 is enabled to avoid the concavity from being formed at

the circumferential portion thereof, unevenness of the upper surface of the chuck table **14** may be reduced. Therefore, intervention of grinding debris between the holder surface **421** and the wafer W due to such unevenness of the upper surface of the chuck table **14** may be avoided, and the wafer W being ground may be prevented from being damaged.

[0069] Furthermore, in the first grinding process, the frame **43** having a higher degree of hardness is ground with the first grinding stones **80** having the larger grain diameter; therefore, the time for grinding in the first grinding process may be shortened. Meanwhile, in the second grinding process, the porous member **42** having a lower degree of hardness is ground with the second grinding stones **90** having the smaller grain diameter. Accordingly, the time for grinding the porous member **42** with the second grinding stones **90** may be longer compared to a hypothetical case in which the porous member **42** is ground with the first grinding stones **80**. However, it provides an effect of smoothing the upper surface of the porous member **42** preferably. Optionally, in the first grinding step, the frame **43** may be ground with the second grinding stones **90**, therefore grinding may be done with the second grinding stones **90** in both the first grinding process and the second grinding process. Furthermore, optionally, in the second grinding step, the porous member **42** may be ground with the first grinding stones **80**, therefore grinding may be done with the first grinding stones **80** in both the first grinding process and the second grinding process. Moreover, optionally, the frame **43** may be ground with the second grinding stones **90** in the first grinding process, and the porous member **42** may be ground with the first grinding stones **80** in the second grinding process.

[0070] Furthermore, the preparatory grinding process enables the upper surface of the porous member **42** and the upper surface of the frame **43** to be ground to align with the same conical surface. The preparatory grinding process may be provided to end in a short time as a process of rough grinding, considering that the angles of the upper surfaces of the porous member **42** and the frame **43** are finely adjusted later by performing the first grinding process and the second grinding process. Therefore, unlike the conventional method, the porous member **42** may prevent the ring-shaped concavity from being formed in the circumferential portion thereof. Moreover, optionally, in the self-grinding operation after grinding the wafers W for a predetermined number of times, the first grinding process and the second grinding process may be performed while the preparatory grinding process is omitted.

[0071] Embodiment of the present disclosure may not necessarily be limited to the configuration described above but may be modified in various ways. In the embodiment described above, sizes or forms of the components illustrated in the accompanying drawings are not limited thereto but may be modified optionally within the scope of the effects of the present disclosure. Moreover, the embodiment may be modified optionally without departing from the scope of the object of the present disclosure.

[0072] For example, the tilt-adjusting assembly **36** may not necessarily be limited to the exemplary configuration as illustrated but may be modified in various ways. For example, the tilt-adjusting assembly **36** may have three or more position-adjusting units **38** or may be configured to change the height of the flange **372** not by rotating but by sliding the movable shaft **382** in the Z-axis direction using, for example, leverage.

[0073] For another example of the tilt-adjusting assembly **36**, the first spindle **76** and the second spindle **86** may be tilted with respect to the Z-axis direction so that the inclination between the table spindle **31** and the first spindle **76** and the inclination between the table spindle **31** and the second spindle **86** are relatively changeable. In this configuration, for example, the tilt-adjusting assembly **36** may be configured to have a first tilt-adjusting assembly provided to the first grinding unit **71** and a second tilt-adjusting assembly provided to the second grinding unit **72**. The first tilt-adjusting assembly may have the axis C**21** of the first spindle **76** inclining with respect to the Z-axis direction so that the relative inclination between the axis C**21** of the first spindle **76** and the axis C**1** of the chuck table **14** forms the first angle $\theta 1$. The second tilt-adjusting assembly may have

the axis C22 of the second spindle **86** inclining with respect to the Z-axis direction so that the relative inclination between the axis C22 of the second spindle **86** and the axis C1 of the chuck table **14** forms the second angle $\theta 2$.

[0074] For another example, in the first and second grinding processes to form the holder surface **421** in the present embodiment, either one of the first grinding unit **71** or the second grinding unit **72** may be used. In other words, the first grinding process may be performed with the first grinding stones **80**, and the first grinding stones **80** may be replaced with the second grinding stones **90** for performing the second grinding process thereafter. Alternatively, the first grinding process may be performed with the second grinding stones **90**, and the second grinding stones **90** may be replaced with the first grinding stones **80** for performing the second grinding process thereafter.

INDUSTRIAL APPLICABILITY

[0075] As explained above, the present disclosure provides the effect such that a ring-shaped concavity may be prevented from being formed on a circumferential portion of a porous member during self-grinding of a chuck table.

Claims

1. A method for forming a holder surface of a chuck table to suction and hold a wafer thereon by rotating a spindle, to which an annular grinding stone is attached, and grinding an upper surface of the chuck table with the grinding stone rotated along with the spindle, the chuck table being composed of a porous member and a frame having a dense body in an annular shape surrounding an outer circumference of the porous member, the method comprising: a first grinding process including tilting an axis of the chuck table with respect to an axis of the spindle relatively at a predetermined first angle and grinding an entire upper surface of the frame with the grinding stone; and a second grinding process including tilting the axis of the chuck table with respect to the axis of the spindle relatively at a second angle, the second angle being smaller than the first angle, and grinding an entire upper surface of the porous member with the grinding stone.
2. The method according to claim 1, wherein a grain diameter of the grinding stone to be used in the second grinding process is smaller than a grain diameter of the grinding stone used in the first grinding process.
3. A grinding apparatus configured to form the holder surface of the chuck table in the method according to claim 1, comprising: the chuck table configured to hold the wafer; a table spindle configured to rotate the chuck table; a grinding unit configured to grind the wafer held on the chuck table with the grinding stone attached to the spindle; a tilt-adjusting assembly configured to adjust relative inclination between the table spindle and the spindle to one of the first angle or the second angle; and a controller configured to control the tilt-adjusting assembly to form the first angle for grinding the frame with the grinding stone and to form the second angle for grinding the porous member with the grinding stone.
4. A grinding apparatus configured to form the holder surface of the chuck table by rotating the spindle, the spindle including a first spindle and a second spindle, and by grinding the upper surface of the chuck table with the grinding stone, the grinding stone including a first grinding stone and a second grinding stone, in the method according to claim 2, the grinding apparatus comprising: the chuck table configured to hold the wafer; a table spindle configured to rotate the chuck table; a first grinding unit configured to grind the wafer held on the chuck table with the first grinding stone attached to the first spindle; a second grinding unit configured to grind the wafer held on the chuck table with the second grinding stone, of which grain diameter is smaller than a grain diameter of the first grinding stone, the second grinding stone being attached to the second spindle; a tilt-adjusting assembly configured to adjust relative inclination between the table spindle and the first spindle to the first angle and relative inclination between the table spindle and the second spindle to the second angle; and a controller configured to control the tilt-adjusting

assembly to form the first angle for grinding the frame with the first grinding stone and to form the second angle for grinding the porous member with the second grinding stone.
